

AYMANE AIT DADS

Applied AI Engineering Intern | End-to-End ML Pipelines · Cloud & Data Engineering · AI Technologies
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PROFESSIONAL SUMMARY

Data Science engineering student who built and deployed an end-to-end ML pipeline on **100K+ fiber-optic network records**, automating feature engineering and model evaluation to cut manual analysis time by ~60% at Orange Maroc. Ranked top 10% on a \$106K+ Kaggle competition leaderboard (0.80/1.0) fine-tuning a 30B Mamba-Transformer MoE with LoRA on novel GPU hardware, demonstrating applied AI engineering under real production constraints. Brings hands-on experience in cloud platforms, databases, data engineering, Python, SQL, PyTorch, and numerical methods directly aligned with GE Aerospace requirements. Available for internship immediately, based in Sophia Antipolis, France, with valid European study authorization and full work eligibility.

CORE COMPETENCIES

Applied AI Engineering | Cloud Platforms | Data Engineering | Numerical Methods | End-to-End ML Pipelines | AI Technologies | Web & Database Design | Real-World Engineering Problems

WORK EXPERIENCE

Orange Maroc Jul 2024 - Aug 2024
Data Science Intern - Casablanca, Morocco

- Engineered an end-to-end ML pipeline processing **100K+ fiber-optic network records** (upload speed, latency, jitter) using Random Forest and K-Means to classify network quality across thousands of nodes.
- Developed a clustering pipeline mapping customer performance profiles to **5 commercial service tiers**, with output adopted directly by the network optimization team for production decision-making.
- Reduced manual analysis time by **~60%** through automated feature engineering, cross-validated model evaluation, and reproducible data workflows built in Python and Scikit-learn.
- Delivered stakeholder presentations translating model outputs and ensemble results into actionable maintenance recommendations, visualized via **Power BI** dashboards for non-technical audiences.

PROJECTS

NVIDIA Nemotron Reasoning Challenge Kaggle Competition · In Progress, June 2026 · Top 10% Public Leaderboard

- Achieved **top 10% public leaderboard (0.80/1.0)** in a \$106K+ prize-pool competition by fine-tuning a 30B Mamba-Transformer MoE with LoRA ($r=32$, ~888M trainable parameters) on 7,828 chain-of-thought reasoning examples using SFTTrainer with completion-only loss.
- Resolved **5 Blackwell GPU (sm_120) incompatibilities** in strict order - including monkey-patched allocator warmup, stubbed Mamba/Cutlass imports, and a pure-PyTorch RMSNorm fallback - and conducted 20+ single-variable ablations confirming packing and synthetic data as performance-degrading factors.

PyTorch · Unsloth · PEFT · Hugging Face Transformers · vLLM · Triton

AI-Generated Image Detection EURECOM ImSecu + NTIRE 2026 @ CVPR · 2025 · Ranked 1st in Class

- Ranked **1st in EURECOM class leaderboard (0.791 AUC)** by fine-tuning CLIP ViT-L/14 (428M params) with a custom MLP head and dual learning rates, then submitted independently to the NTIRE 2026 @ CVPR CodaBench international benchmark.
- Built a 3-model ensemble weighted by validation AUC with **10-view test-time augmentation** (flips, crops, blur, resizes) trained on 250K images across 25+ generator types using FP16 mixed precision on T4 GPU.

PyTorch · OpenCLIP · ViT · AdamW · Kaggle GPU

EDUCATION

Engineering Degree in Data Science (Master-level) - EURECOM Sep 2023 - Present

Relevant coursework: Machine Learning, Deep Learning, Computer Vision, NLP, Cloud Computing, Statistics, Image Security, Web Applications

CPGE - Mathematics & Physics - Ibn Timiya 2021 - 2023

SKILLS

ML Frameworks & Engineering: PyTorch | Scikit-learn | Hugging Face Transformers | PEFT (LoRA, QLoRA) | Unsloth | TRL | OpenCLIP

Data Engineering & Cloud: Python | SQL | Pandas | NumPy | Supabase | PostgreSQL | REST APIs | Docker | Power BI

AI Technologies & MLOps: LLM Fine-tuning | vLLM | RAG | LangChain | FP16 Mixed Precision | CUDA Debugging | Git | Kaggle GPU